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MATHEMATICS
EXPLORATION PROBLEMS

Name : _____ **Index Number** : _____

Country : _____



18th International Mathematics and Science Olympiad
Indonesia
16 December 2021

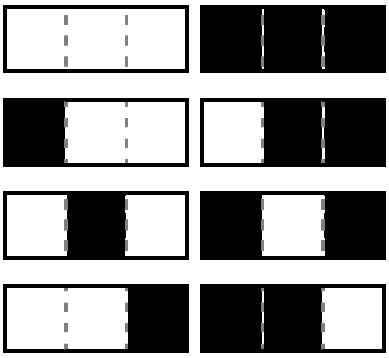
Instructions:

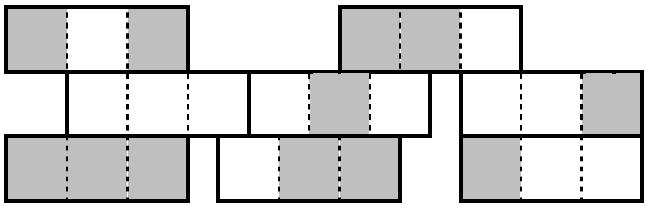
1. Write your name, country and index number on every page of the Answer Sheet.
2. Write your answers only in the Answer Sheet.
3. Answer all questions in Arabic Numerals or English.
4. There are 6 questions in this paper.
5. Each question is worth 6 marks and partial credit may be awarded.
6. You have 120 minutes to complete this paper.
7. You are provided with some manipulatives for exploration on some questions.
8. Use black pen or blue pen or pencil to write your answer.

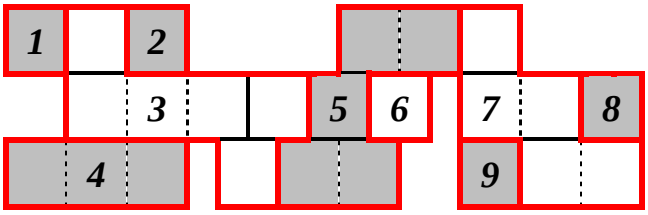
Do not turn over this page until you are told to do so.

EXPLORATION PROBLEMS

1. There are eight different 1×3 tiles as shown on the right. The diagram below on the left shows one arrangement of these tiles that are placed on a table. Every connected region of squares of the same colour scores 1 point, so that the score for this arrangement is 9, from 6 black regions and 3 white regions as shown below on the right. The tiles are now re-arranged in order to minimize the score. They may not be rotated or reflected, but they may be moved in any direction by any distance which needs not be integral.
What is the minimum score that can be achieved? Draw one example.







2. (a) There are seven "2"s that are written consecutively: 2222222. Nala will add exactly two "+" operators and one "–" operator in between some of these "2"s such that the result of the expression is a positive multiple of 3. For example, $222+2+2-22=204$ is divisible by 3. In how many ways can Nala do this? (2 points)

   (b) Now, consider ten "2"s that are written consecutively: 2222222222. Nala will add exactly five "+" operators and one "–" operator in between some of these "2"s such that the result of the expression is a positive multiple of 15. For example, $222+2+2+2+2+2-22=210$ is divisible by 15. In how many ways can Nala do this? (4 points)

3. (a) In the diagram on the right, there is a 3×3 square where each of its unit squares contains the digit from 1 to 9. How many different rectangles (squares included) are there such that the sum of all numbers inside it is a multiple of 3? (2 points)

1	4	7
2	5	8
3	6	9

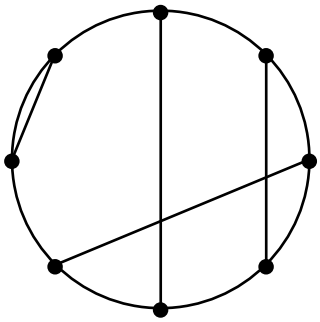
   (b) In the diagram on the right, there is a 5×5 square where each of its unit squares contains the numbers from 1 to 25. How many different rectangles (squares included) are there such that the sum of all numbers inside it is a multiple of 5? (4 points)

1	6	11	16	21
2	7	12	17	22
3	8	13	18	23
4	9	14	19	24
5	10	15	20	25

4. The IQ (intelligence quotient) of a country is defined as the average IQ of its entire population. During a migration period, the total population and individual IQs remain constant throughout.
   (a) A group of people from country A emigrates to country B. Is it possible that the average IQs of both countries have increased? (1 point)
   (b) After this, a group of people from country B emigrates to country A. Is it possible that the average IQs of both countries will increase again? (2 points)
   (c) A group of people from country A emigrates to country B and a group of people from country B emigrates to country C. Is it possible that the average IQs of all three countries have increased? (3 points)

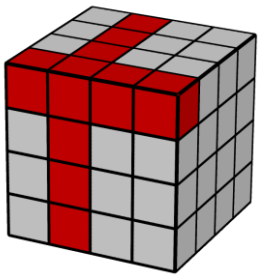
   If your answer is yes, provide an example. If your answer is no, explain why.

5. There are n points that are placed equally distanced on a circle.
   We wish to connect the points in pairs such that all segments formed have different lengths.
   For example, if $n=8$, then we can connect the points as shown in the diagram on the right:
   Is this possible if
   (a) $n=10$ (1 point);
   (b) $n=12$ (2 points);
   (c) $n=20$ (3 points)?

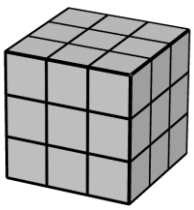
   

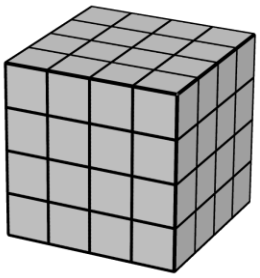
   If your answer is yes, please provide an example.
   If your answer is no, explain why.

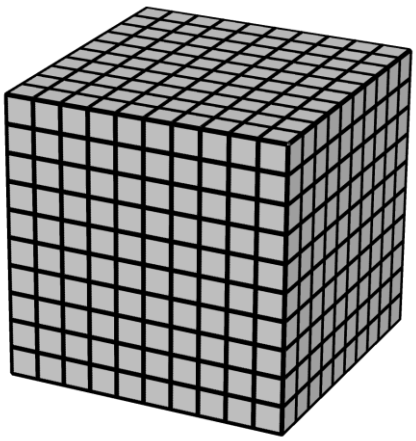
6. In a $n\times n\times n$ block, choose any unit cube and remove it along with every other unit cube in the same row, the same column or the same stack.

   Determine the minimum number of unit cubes we must choose in order to remove all n^3 unit cubes, if
   (a) $n=3$ (1 point); (b) $n=4$ (2 points); (c) $n=10$ (3 points).

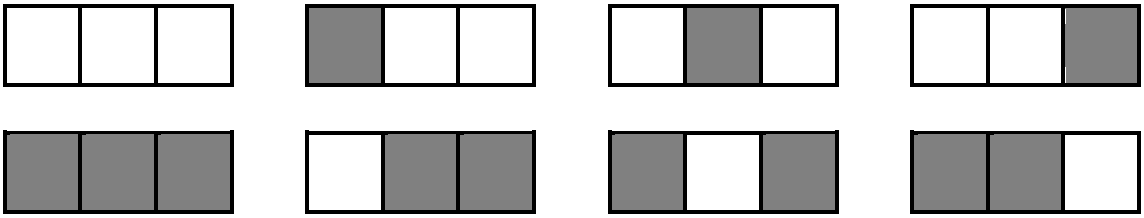
   

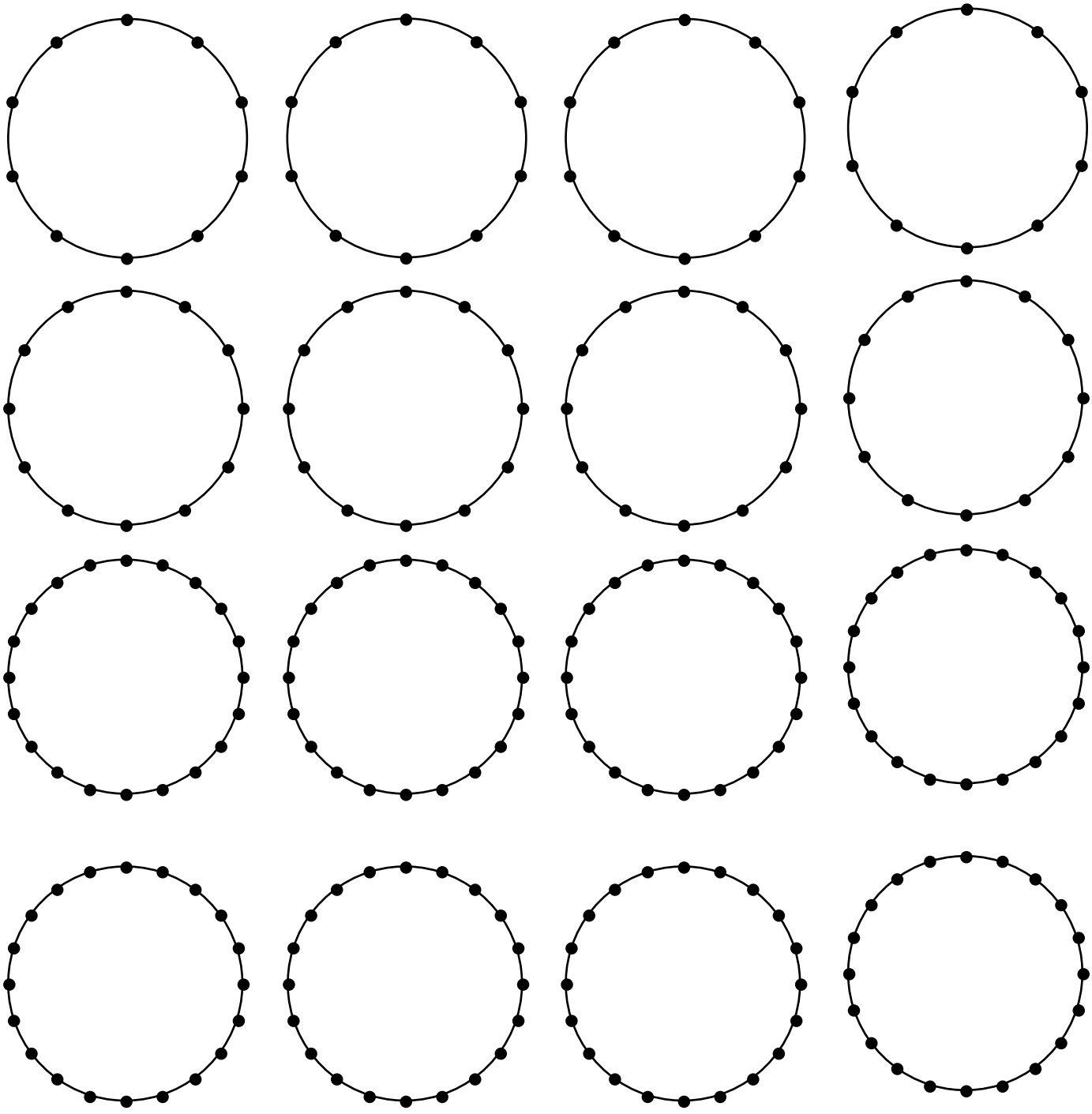
   

For Sketch Use

(1)

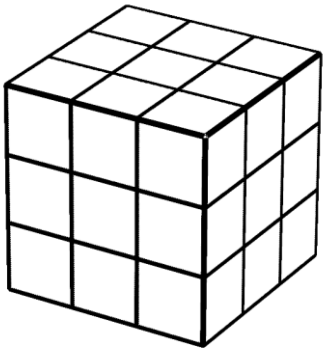


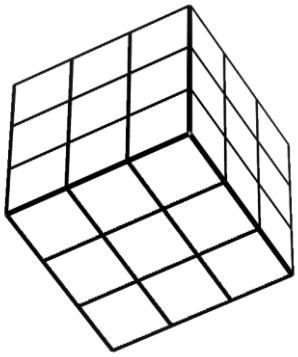
(5)

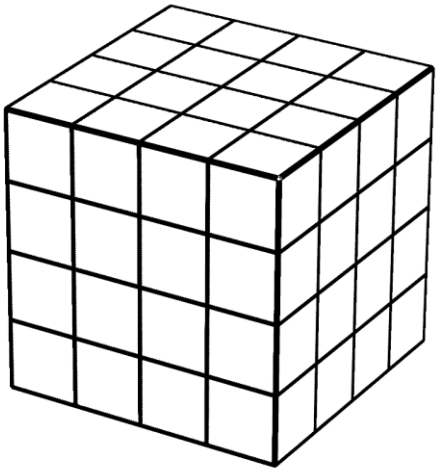


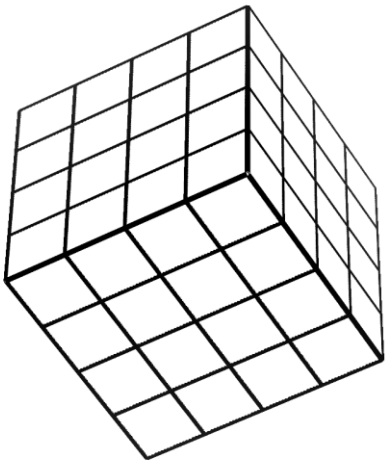
For Sketch Use

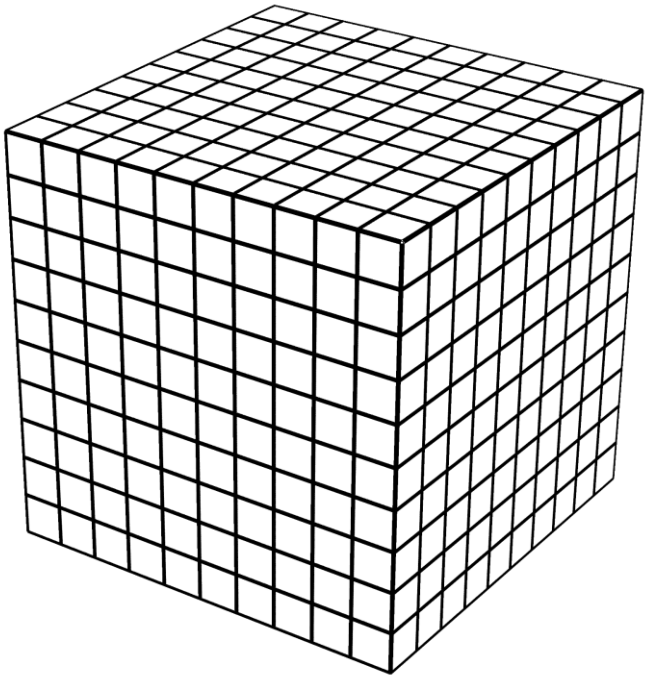
(6)

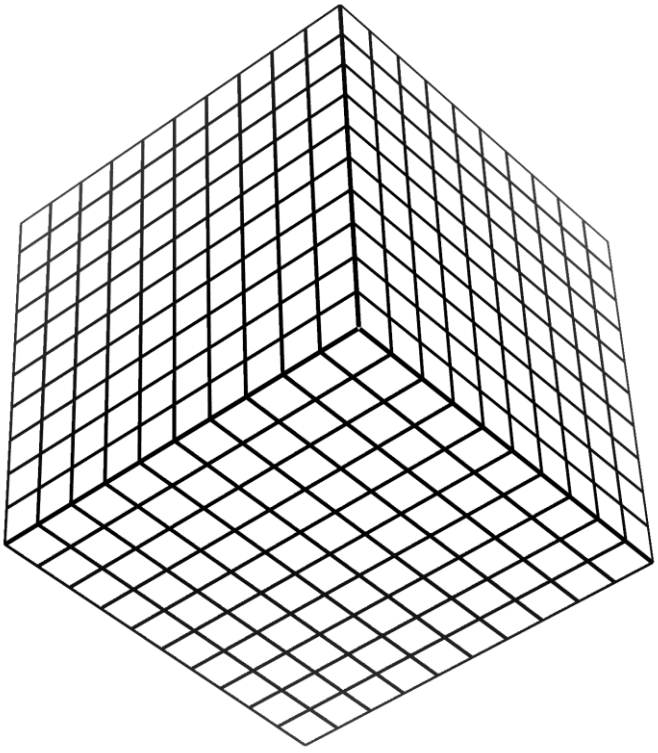












MATHEMATICS
EXPLORATION PROBLEMS
ANSWER SHEET

Name : _____ **Index Number** : _____

Country : _____

18th International Mathematics and Science Olympiad

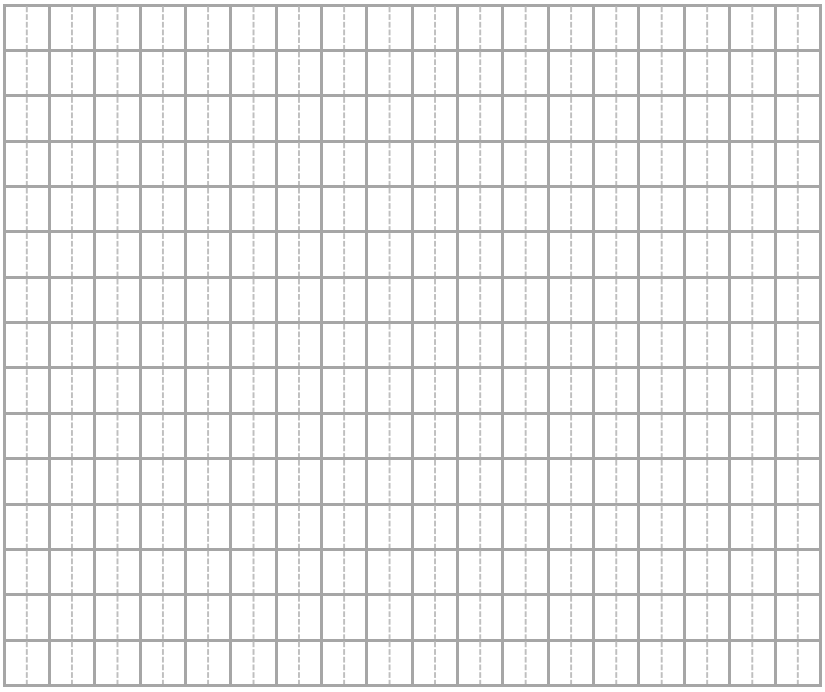
Indonesia

16 December 2021

EXPLORATION PROBLEMS
ANSWER SHEET

Country**:** _____ Name**:** _____ Index Number : _____

1. The minimum score is _____ points as shown in the diagram below :



2. (a) Nala has _____ ways to add 2's "+" operator and 1 "–" operator in between two of 2, so, the results of final operation is positive integer and divisible by 3. **(no partial score.)**

   (b) Nala has _____ ways. ways to add 5's "+" operator and 1 "–" operator in between two of 2, so, the results of final operation is positive integer and divisible by 15. **(no partial score.)**

3. (a) There are _____ rectangles that the sum of all numbers inside the rectangle is a multiple of 3. **(no partial score.)**

   (b) There are _____ rectangles that the sum of all numbers inside the rectangle is a multiple of 5. **(no partial score.)**

EXPLORATION PROBLEMS
ANSWER SHEET

Country**:** _____ Name**:** _____ Index Number : _____

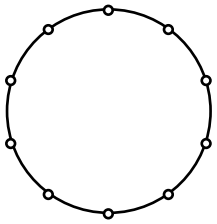
4. (a) ☐ Possible ☐ Impossible
The reason is _____

(b) ☐ Possible ☐ Impossible
The reason is _____

(c) ☐ Possible ☐ Impossible
The reason is _____

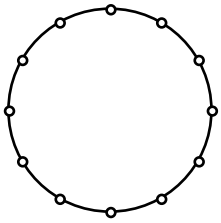
If the answer is just possible or impossible, without any example or reasoning, 0 marks

5. (a) ☐ Possible ☐ Impossible



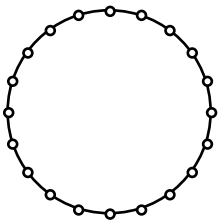
The reason is _____

(b) ☐ Possible ☐ Impossible



The reason is _____

(c) ☐ Possible ☐ Impossible



The reason is _____

If the answer is just possible or impossible, without any example or reasoning, 0 marks

6. (a) If $n = 3$, then the minimum number of unit cubes is _____.
(no partial score.)

(b) If $n = 4$, then the minimum number of unit cubes is _____.
(no partial score.)

(c) If $n = 10$, then the minimum number of unit cubes is _____.
(no partial score.)